

CADRE ERP PLATFORM

Architecture & Technical Specification

Complete System Design · Database Schema · Role Access · Module Breakdown

DOCUMENT INFORMATION

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1. Executive Summary

This document defines the complete software architecture, module breakdown, entity relationship model, and role-based access framework for the Cadre Custom ERP Platform. The platform is purpose-built ground-up for Cadre, comprising five tightly integrated modules designed to eliminate manual effort, reduce errors, and provide full visibility into every client engagement.

The system serves five internal staff roles (Super Admin, Sales, CSR, Operations, Accounts) through dedicated portals, plus an external Client Portal. An AI Tools layer — beginning with the Bank Statement Report Generator — extends the platform with intelligent automation powered by the Claude API.



2. System Overview

The Cadre ERP is a web-based multi-portal application with a single backend API serving multiple frontend portals. Each role accesses the system through its own portal URL, and all permissions are enforced at both the API and database levels.

2.1 High-Level Architecture

- Technology stack recommendation:
- Backend API: Laravel (PHP) or Node.js (NestJS) — REST + WebSockets
 - Frontend: React.js — separate builds per portal
 - Database: PostgreSQL with row-level security
 - File Storage: AWS S3 or Cloudflare R2
 - WhatsApp: Meta Cloud API (Business API) or Twilio
 - AI Layer: Anthropic Claude API (claude-sonnet-4-20250514)
 - Queue System: Redis + Bull for background jobs
 - Hosting: Docker containers on VPS or managed cloud

2.2 Portal Architecture

Each portal is a separate frontend build with its own domain/subdomain:

Portal	URL Pattern	Access Scope
Admin Portal	admin.cadre.app	Full system - Super Admin only
Staff Portal	app.cadre.app	Sales, CSR, Operations, Accounts
Client Portal	client.cadre.app	External clients - projects & invoices
AI Tools	tools.cadre.app	Staff access - AI-powered tools

3. Role-Based Access Control

The system enforces permissions at three layers: frontend route guards, API middleware, and database row-level policies. Each role has a defined set of allowed actions — no role can access functionality outside its permission matrix.

Role	Portal Access & Permissions
Super Admin	Full system access, all modules, configuration, services, payroll, commissions, user management
Sales	Add clients, leads management, follow-ups, create invoices, create projects, view own commission reports
CSR	Document proof check, customer support, view client profiles, communication logs
Operations	Full project management, step creation, task assignment, update status, client WhatsApp notifications
Accounts	Attendance & payroll, project recoveries, expense entry, payment aging, financial reports
Client Portal	View projects & steps, upload documents per step, view and pay invoices, view credential tabs

3.1 Permission Enforcement Layers

- Layer 1 — Frontend: Route guards prevent navigation to unauthorized pages
- Layer 2 — API Middleware: JWT token + role check on every endpoint
- Layer 3 — Database: Row-level security policies in PostgreSQL
- Audit: Every action is written to the activity_logs table with user, IP, entity, and timestamp

4. Module Architecture

01

Client Management Module

Onboarding · Portal · Credentials · Documents

The Client Management module is the central hub for all client-facing data. Every entity in the system traces back to a client record.

Key Features

- Client onboarding with CNIC, WhatsApp number, email (optional), and full name
- Automatic portal credential generation — sent to client via WhatsApp and email
- Per-client commission rate configured by Super Admin (used in commission calculation)
- Credentials tab — encrypted storage of third-party login credentials and notes per client
- Document repository — all documents uploaded for or by client, linked to steps or general
- Invoice tracking — status, payment history, and outstanding amounts per client

Commission Rate Logic

Commission is calculated exclusively on service charges within an invoice — not on challans, government fees, or other third-party charges. The commission rate is stored as a snapshot on the commission record at time of invoice payment, so historical rates are preserved even if the client rate changes.

02

Role-Based Employee Access System

Permissions · Audit Trail · Activity Logs

Controls all internal staff access. Permissions are stored as structured JSON in the roles table, enabling fine-grained control without schema changes.

- Granular permission levels: Admin, Manager, Staff, Viewer per module
- Each employee sees only what is relevant to their role
- Activity logs and audit trail for accountability — immutable once written

- Task assignment and workload visibility

03

AI-Powered WhatsApp Agent Module

Lead Capture · CRM Pipeline · Escalation

The WhatsApp agent is powered by a custom knowledgebase trained on Cadre's services, pricing, and FAQs. It handles inbound inquiries automatically and escalates to human agents when needed.

- Custom knowledgebase trained on services, FAQs, and pricing
- Automated lead capture from WhatsApp conversations
- Lead qualification, follow-up reminders, and CRM pipeline management
- Seamless escalation to a human agent when required
- Analytics dashboard — messages received, leads captured, conversions

04

Project Management Module

Steps · Document Forms · WhatsApp Alerts · Charges

The most complex module. Every project is linked to a service and progresses through employee-defined steps. Clients receive WhatsApp notifications at each step and can upload required documents through the client portal.

Project Creation Flow

- Employee selects service → client → creates project
- Employee adds steps manually — unlimited steps per project
- Each step has: title, description, optional document form, duration in days, optional charge
- If step requires documents, employee defines the form fields (field name, type, required)
- Duration timer starts when step becomes active — alert fired if not completed in time
- WhatsApp notification sent to client automatically on each new step
- Step-level charges (e.g. Challan) are added to the project invoice automatically
- Employee marks project complete only when all steps are done

Document Form Builder

The doc_form_fields column stores a JSON array of field definitions. Example structure:

```
[ { "key": "cnic_front", "label": "CNIC Front Side", "type": "file", "required": true },
  { "key": "bank_statement", "label": "Bank Statement PDF", "type": "file", "required": true },
  { "key": "ntn", "label": "NTN Number", "type": "text", "required": false } ]
```

05

Reporting & Finance Module

Revenue · Commissions · Payroll · Expenses

Provides complete financial visibility for the Accounts team and management. All reports are exportable as PDF or Excel.

- Revenue dashboard — invoices raised, collected, and outstanding
- Commission reports — calculated on service charges only, not challans
- Project profitability and team performance reports
- Attendance tracking and monthly payroll processing
- Expense entry with receipt attachments per project or general
- Payment aging — overdue invoices by duration
- Recovery tracking per project

5. AI Tools — App Section

The AI Tools section is a dedicated area of the platform where staff can use Claude-powered tools to automate complex document processing tasks. Each tool runs as an async background job and stores results for later retrieval and download.

T1

Bank Statement Report Generator

Claude API · PDF Processing · Structured Report

Staff uploads a bank statement PDF. The system sends it to the Claude API for analysis and returns a structured 2-page report that can be viewed in-app or downloaded as PDF.

Report Output Fields

#	Field	Description
1	Total Credit	Sum of all credit transactions
2	Total Debit	Sum of all debit transactions
3	Opening & Closing Balance	Account balance at start and end of statement period
4	Unusual Transactions	Flagged outliers, large amounts, or irregular patterns
5	Bank-to-Bank Online Transfers	Total amount of electronic inter-bank transfers
6	Cash Deposits	Total cash deposited over the statement period
7	Cheque Deposits	Total amount deposited via cheque
8	Remittances	Total inward/outward remittance amounts
9	Banking Profits & Tax Deducted	Profit on savings/TDRs and withholding tax deducted
10	General Summary	AI-generated narrative overview of account activity

Processing Flow

- Staff uploads PDF → job queued in `ai_tool_jobs` table
- Worker sends PDF (base64) to Claude API with structured extraction prompt
- Claude returns JSON with all 10 fields
- Results stored in `statement_report_results` table
- PDF report generated (2 pages) and stored in file storage
- Staff can view report inline or download PDF

6. Entity Relationship Diagram — Core Entities

All tables use UUID primary keys. Timestamps are stored in UTC. The database is PostgreSQL with row-level security enabled on sensitive tables.

users — Internal staff accounts

Field	Type	Notes
PK id	uuid	Primary key
name	varchar(150)	
email	varchar(255)	Unique
password_hash	varchar	bcrypt

FK role_id	uuid	→ roles
is_active	boolean	Soft disable
created_at	timestamp	

roles — Permission definitions

Field	Type	Notes
PK id	uuid	
name	varchar(50)	super_admin, sales, csr, operations, accounts
permissions	jsonb	Structured permission map per module
created_at	timestamp	

clients — Client records

Field	Type	Notes
PK id	uuid	
full_name	varchar(200)	
cnic	varchar(15)	Unique
whatsapp_number	varchar(20)	
email	varchar(255)	Nullable
commission_rate	decimal(5,2)	Set by Super Admin
portal_username	varchar(100)	Auto-generated
portal_password_hash	varchar	
FK created_by_user_id	uuid	→ users (sales)
is_active	boolean	
created_at	timestamp	

client_credentials — Saved credentials per client

Field	Type	Notes
PK id	uuid	
FK client_id	uuid	→ clients
label	varchar(200)	e.g. FBR Portal, SECP Login
username	varchar(255)	AES-256 encrypted
password	text	AES-256 encrypted
notes	text	Nullable
created_at	timestamp	

client_documents — General document store

Field	Type	Notes
PK id	uuid	
FK client_id	uuid	→ clients
label	varchar(200)	
file_url	text	S3/R2 URL

uploaded_by	uuid	→ users or null (client upload)
created_at	timestamp	

7. Entity Relationship Diagram — Leads & Services

leads — Inbound lead tracking

Field	Type	Notes
PK id	uuid	
name	varchar(200)	
whatsapp_number	varchar(20)	
source	enum	whatsapp_bot, manual, referral, web
status	enum	new, contacted, qualified, converted, lost
FK assigned_to_user_id	uuid	→ users (sales)
FK converted_to_client_id	uuid	→ clients (nullable)
notes	text	
follow_up_at	timestamp	Nullable
created_at	timestamp	

services — Service catalog

Field	Type	Notes
PK id	uuid	
name	varchar(200)	e.g. Tax Filing, Business Registration
description	text	Nullable
default_price	decimal(12,2)	Nullable — starting price guide
is_active	boolean	
created_at	timestamp	

8. Entity Relationship Diagram — Projects & Finance

projects — Client projects

Field	Type	Notes
PK id	uuid	
FK client_id	uuid	→ clients
FK service_id	uuid	→ services
FK assigned_to_user_id	uuid	→ users (operations)
FK created_by_user_id	uuid	→ users
title	varchar(300)	
status	enum	active, on_hold, completed, cancelled
started_at	timestamp	

completed_at	timestamp	Nullable
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project_steps — Steps within a project

Field	Type	Notes
PK id	uuid	
FK project_id	uuid	→ projects
title	varchar(300)	
description	text	
status	enum	pending, active, completed
doc_form_fields	jsonb	Array of field definitions (nullable)
duration_days	integer	Expected completion time
charge_amount	decimal(12,2)	Nullable — challan or step charge
charge_type	enum	service_charge, challan, other
sort_order	integer	Ordering of steps
activated_at	timestamp	Nullable
completed_at	timestamp	Nullable

step_doc_submissions — Client document uploads per step

Field	Type	Notes
PK id	uuid	
FK step_id	uuid	→ project_steps
field_key	varchar(100)	Matches key in doc_form_fields
file_url	text	S3/R2 URL
uploaded_at	timestamp	

invoices — Client invoices

Field	Type	Notes
PK id	uuid	
invoice_number	varchar(50)	Unique, auto-generated
FK client_id	uuid	→ clients
FK project_id	uuid	→ projects (nullable)
FK created_by_user_id	uuid	→ users (sales)
status	enum	draft, sent, partial, paid, overdue
total_amount	decimal(12,2)	service_charges + other_charges
service_charges_total	decimal(12,2)	Commission base — service only
other_charges_total	decimal(12,2)	Challans, govt fees (non-commissionable)
amount_paid	decimal(12,2)	Running total from payments
due_date	date	
issued_at	timestamp	

invoice_line_items — Individual line items per invoice

Field	Type	Notes
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PK id	uuid	
FK invoice_id	uuid	→ invoices
FK step_id	uuid	→ project_steps (nullable)
description	varchar(500)	
item_type	enum	service_charge, challan, other
amount	decimal(12,2)	
is_commissionable	boolean	True for service_charge only

commissions — Salesperson commission records

Field	Type	Notes
PK id	uuid	
FK sales_user_id	uuid	→ users (sales role)
FK client_id	uuid	→ clients
FK invoice_id	uuid	→ invoices
commission_rate	decimal(5,2)	Snapshot at time of payment
base_amount	decimal(12,2)	service_charges_total only
commission_amount	decimal(12,2)	base_amount x rate / 100
status	enum	pending, approved, paid
generated_at	timestamp	Auto on invoice paid

payments — Partial/full invoice payments

Field	Type	Notes
PK id	uuid	
FK invoice_id	uuid	→ invoices
amount	decimal(12,2)	
method	enum	bank_transfer, cash, cheque, online
reference_number	varchar(100)	Nullable
FK recorded_by_user_id	uuid	→ users (accounts)
paid_at	timestamp	

attendance — Daily employee attendance

Field	Type	Notes
PK id	uuid	
FK user_id	uuid	→ users
date	date	
check_in	time	Nullable
check_out	time	Nullable
status	enum	present, absent, half_day, leave
notes	text	Nullable

payroll — Monthly payroll records

Field	Type	Notes
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PK id	uuid	
FK user_id	uuid	→ users
month	date	First day of the month
base_salary	decimal(12,2)	
deductions	decimal(12,2)	Absences, advances
bonuses	decimal(12,2)	Including commissions if applicable
net_salary	decimal(12,2)	base - deductions + bonuses
status	enum	draft, approved, paid
paid_at	timestamp	Nullable

9. Entity Relationship Diagram — AI Tools & WhatsApp

ai_tool_jobs — AI job queue

Field	Type	Notes
PK id	uuid	
tool_type	enum	statement_report (extensible)
FK requested_by_user_id	uuid	→ users
input_file_url	text	Uploaded PDF URL
status	enum	queued, processing, done, failed
output_file_url	text	Generated report URL (nullable)
error_message	text	Nullable
created_at	timestamp	
completed_at	timestamp	Nullable

statement_report_results — Parsed bank statement output

Field	Type	Notes
PK id	uuid	
FK job_id	uuid	→ ai_tool_jobs
account_title	varchar(200)	
account_number	varchar(50)	
opening_balance	decimal(15,2)	
closing_balance	decimal(15,2)	
total_credits	decimal(15,2)	
total_debits	decimal(15,2)	
bank_to_bank_transfers	decimal(15,2)	Online inter-bank
cash_deposits	decimal(15,2)	
cheque_deposits	decimal(15,2)	
remittances	decimal(15,2)	
banking_profits	decimal(15,2)	Savings/TDR profit
tax_deducted_on_profit	decimal(15,2)	Withholding tax

unusual_transactions	jsonb	Array of flagged items
general_summary	text	AI narrative overview
report_pdf_url	text	Generated 2-page report

wa_conversations — WhatsApp conversation threads

Field	Type	Notes
PK id	uuid	
wa_phone_number	varchar(20)	
FK lead_id	uuid	→ leads (nullable)
FK client_id	uuid	→ clients (nullable)
status	enum	ai_handling, escalated, closed
escalated_to_user_id	uuid	→ users (nullable)
started_at	timestamp	
last_message_at	timestamp	

wa_messages — Individual WhatsApp messages

Field	Type	Notes
PK id	uuid	
FK conversation_id	uuid	→ wa_conversations
direction	enum	inbound, outbound
sender_type	enum	contact, ai_agent, staff
body	text	
wa_message_id	varchar(100)	Unique — Meta API ID
sent_at	timestamp	

10. Entity Relationships Summary

The table below summarizes all primary foreign key relationships in the system.

From Entity	Cardinality	To Entity	Description
users	N → 1	roles	Each staff user has one role
clients	N → 1	users (sales)	Created_by tracks salesperson
clients	1 → N	client_credentials	Many saved credentials per client
clients	1 → N	client_documents	Document repository per client
leads	N → 1	clients	Lead converts to client
projects	N → 1	clients	Client can have many projects
projects	N → 1	services	Each project linked to one service
projects	1 → N	project_steps	Unlimited steps per project
project_steps	1 → N	step_doc_submissions	Per-field uploads from client
project_steps	1 → N	invoice_line_items	Step charges flow into invoice
invoices	N → 1	clients	Invoice belongs to client

invoices	1 → N	invoice_line_items	Service vs. challan split here
invoices	1 → 1	commissions	Auto-generated on invoice paid
commissions	N → 1	users (sales)	Rate snapshot from client record
invoices	1 → N	payments	Partial payment support
users	1 → N	attendance	Daily records per employee
users	1 → N	payroll	Monthly payroll per employee
ai_tool_jobs	1 → 1	statement_report_results	Parsed output stored separately
wa_conversations	1 → N	wa_messages	Full message thread
wa_conversations	N → 1	leads	AI captures lead from conversation

11. Commission Calculation Logic

Commission calculation is a critical business rule. The following defines exactly how commissions are computed and what is included or excluded.

11.1 What is Commissionable

- INCLUDED: Service charges — the core fee charged for the service (Tax Filing fee, Business Registration fee, etc.)
- EXCLUDED: Challans — government payment challans paid on behalf of the client
- EXCLUDED: Court fees, filing fees, third-party costs
- EXCLUDED: Any invoice_line_item where is_commissionable = false

11.2 Calculation Formula

```
commission_amount = service_charges_total x (commission_rate / 100)
```

Where commission_rate is the rate set on the client record by the admin at time of invoice payment. The rate is snapshotted on the commission record to preserve historical accuracy.

11.3 Commission Trigger

- Commission record is auto-generated when invoice status changes to 'paid'
- Status starts as 'pending', moves to 'approved' by accounts, then 'paid' when processed in payroll
- Commission can be viewed by the salesperson in their portal (read-only)

12. Security & Integration Notes

12.1 Security

- Authentication: JWT tokens with short expiry (15 min access + refresh token)
- Client credentials stored AES-256 encrypted at rest
- All API endpoints require role validation middleware
- Activity logs are append-only — no updates or deletes
- File uploads scanned for malware before storage

- HTTPS enforced everywhere — no mixed content

12.2 WhatsApp Integration

- Provider: Meta Cloud API (recommended) or Twilio WhatsApp Business
- Webhook receives inbound messages, routed to AI agent or escalated staff
- Outbound notifications triggered by: new project step, step completion, invoice sent, project completed
- All messages logged in wa_messages for full audit trail

12.3 AI / Claude API Integration

- Model: claude-sonnet-4-20250514
- Bank statements sent as base64-encoded documents in the API request
- Prompt engineered to return structured JSON matching statement_report_results schema
- Results validated before storage — job marked failed if parsing fails
- API key stored as environment variable, never in codebase

13. Next Steps

#	Action Item	Owner
1	Review and sign off on this architecture document	Client + Dev Team
2	Confirm tech stack selection (Laravel vs Node.js)	Dev Lead
3	Select WhatsApp Business API provider	Client
4	Set up cloud infrastructure (hosting, DB, storage)	Dev Team
5	Begin database schema implementation in PostgreSQL	Backend Dev
6	Design wireframes for each portal (6 portals)	UI/UX Designer
7	Integrate Meta Cloud API for WhatsApp	Backend Dev
8	Build AI Statement Report tool (Claude API)	AI Dev
9	UAT — client reviews each module in staging	Client + QA
10	Staff training and go-live	Project Manager